

# KAJAL SANWAL

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## PROFESSIONAL SUMMARY

Software Engineer | Software Development Engineer | Full-Stack | Backend

Experience building scalable, high-performance applications across full-stack web development and embedded systems. Proficient in React, Next.js, Node.js, PostgreSQL, and REST API development. Strong foundation in C/C++, firmware development, and hardware-software integration. Familiar with microservices architecture, distributed systems basics, and performance optimization. Pursuing BS in Electronic Systems at IIT Madras.

## EDUCATION

Degree / Board - Institute Year: GPA / %

BS, Electronic Systems - IIT Madras: 2027 (8.32)

## SCHOLASTIC ACHIEVEMENTS

- JEE Main Percentile: 99.1
- LeetCode: Solved 150+ problems (Data Structures & Algorithms); regular participant in weekly and biweekly contests
- Science Olympiad (SOF): Secured olympiad rank 62 for distinctive performance, 2019.
- International Reasoning & Aptitude Olympiad (SilverZone): Achieved Rank 96, 2018.
- 1st Place, State Level Science Drama Contest — State Council of Educational Research and Training (SCERT).

## EXPERIENCE

**Embedded Systems / Firmware Intern — Instrumentation Team** — BluArmor | Aug 2025 – Oct 2025

*Pre-Placement Offer (PPO) extended upon completion*

- Designed automated RF/Bluetooth characterization systems using STM32 firmware, relay matrices, and ADC acquisition, reducing manual testing time by 15%.
- Built Python-based calibration and automation workflows, improving production measurement accuracy by 33%. • Developed Python GUI tools for Bluetooth RSSI testing with automated CSV logging and live visualization, increasing throughput to 500 devices/day.
- Improved RSSI measurement stability by 55% using RF shielding and Exponential Moving Average (EMA) filtering. • Automated headset pairing & validation workflows on Raspberry Pi 5 and nRF52840, enabling parallel multi-device testing.

## PROJECTS & SYSTEM DESIGN

**1. AI-Powered SaaS Platform** | Next.js · Prisma · NeonDB · Clerk · Cloudinary | [Live](#) | [GitHub](#)

- Built scalable multi-tenant SaaS system supporting 50+ concurrent tenants with role-based authentication. • Integrated Cloudinary for optimized media uploads and transformations, reducing media load time by ~40%. • Deployed serverless AI workloads with high-availability auto-scaling on Vercel edge infrastructure.

**2. VibeCode Editor** | React · Zustand · Appwrite · TypeScript | [GitHub](#)

- Developed a high-performance browser-based IDE with virtualized file explorer and live execution playground, achieving sub-50ms re-render latency.
- Managed complex state with Zustand and utilized Appwrite BaaS for secure database schemas and real-time sync.

**3. Fastify Backend & Thumbnailing Service** | Fastify · Node.js Streams · JWT · PostgreSQL | [Live](#) | [GitHub](#)

- Built scalable backend system using Fastify with modular architecture and Node.js streams for efficient file handling up to 100MB.
- Designed JWT authentication and structured logging; average API response time under 80ms. • Implemented RESTful APIs with optimized latency and structured logging.

**4. Full-Stack Auth & Admin Notes** | Node.js · MongoDB · Express · JWT · RBAC | [Live](#) | [GitHub](#) •

- Built JWT + RBAC authentication system deployed to 200+ active IIT Madras users.
- Designed indexed MongoDB schemas reducing query time by ~30%; implemented secure note approval workflows.

## TECHNICAL SKILLS

**Languages:** C/C++, Embedded C, Python, JavaScript, TypeScript, SQL, Linux Shell

**Frontend:** React, Next.js (App Router), HTML, CSS, Tailwind CSS, shadcn/ui

**Backend & DBs:** Node.js, Express, Fastify, REST APIs, MongoDB, PostgreSQL (Neon), Prisma ORM

**Auth & Security:** JWT, RBAC, Clerk, OAuth basics, Secure API Design

**Embedded / HW:** STM32 HAL, FreeRTOS, PWM, Raspberry Pi 5, nRF52840, PCB design, MATLAB, LTSpice

**DevOps / Tools:** Git & GitHub, Vercel, CI/CD basics, Docker (familiar), Postman, Cloudinary, VS Code **Core CS:**

Data Structures & Algorithms, OOP, System Design Fundamentals, Problem Solving **Concepts:** Scalable Systems, Microservices Architecture, Distributed Systems (Basics), Performance Optimization

## POSITIONS OF RESPONSIBILITY

**Deputy Head, Anukriti — Paradox, IIT Madras** *March 2025*

**Teaching Assistant, Embedded C Programming Lab — IIT Madras** *Sept 2025*

**Teaching Assistant, Electronic Systems Thinking & Circuits Lab — IIT Madras** *May – June 2025*

**Teaching Assistant, Embedded C Programming Lab — IIT Madras** *May – June 2025*

**Coordinator, Spark (ES Lab Activities) — IIT Madras** *Jan 2025*

**Group Leader, Kanha House — IIT Madras** *Jan 2024 – Aug 2024*

**Logistic Team Coordinator, Kanha House — IIT Madras** *Oct 2023 – Jul 2024*

Teaching Assistant responsibilities included:

- Mentored 60+ students in embedded C programming, microcontroller debugging, and low-level memory optimization.
- Guided hands-on circuit design and system-level troubleshooting, improving lab performance and debugging efficiency.

## EXTRACURRICULAR ACTIVITIES

- Member, Robosoccer Team — Paradox, IIT Madras (June 2024)
- Member, IITM DIY School-Connect Program — IIT Madras (July 2024 – Jan 2025)
- Gold Medalist, Maths Olympiad (1st rank at School level)
- Won Mathematics Aptitude Test at school level — Army Public School